Quantification: 682_z23
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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| measure | value |
|---|---|
| lacuna count | 14 |
| interior lacuna count | 11 |
| frame-edge lacuna count | 3 |

Interior lacunae only: mean, median and sample SD (n = 11)
----------------------------------------------------------

| measure | kind | mean | median | sd | units |
|---|---|---|---|---|---|
| area_px2 | lacuna shape | 1994.91 | 1519 | 1026.72 | px^2 |
| major_axis_length_px | lacuna shape | 85.7455 | 86.1566 | 22.5145 | px |
| minor_axis_length_px | lacuna shape | 30.4546 | 26.8577 | 8.8339 | px |
| aspect_ratio | lacuna shape | 2.9668 | 3.0363 | 0.8704 | unitless |
| eccentricity | lacuna shape | 0.92 | 0.9442 | 0.0643 | unitless |
| solidity | lacuna shape | 0.8555 | 0.8762 | 0.0782 | unitless |
| roots_count | headline | 10.3636 | 10 | 3.2641 | count |
| ring_length_r30_px | headline | 423.091 | 442 | 90.8366 | px |
| ring_length_r60_px | reported | 989.091 | 998 | 188.498 | px |
| owned_length_px | ownership-dependent | 2954.87 | 2421.54 | 1411.64 | px |
| edge_count | ownership-dependent | 138.273 | 103 | 69.933 | count |
| mean_edge_length_px | ownership-dependent | 21.6005 | 21.4048 | 1.8936 | px |
| perimeter_px | normalised | 221.572 | 210.244 | 51.1926 | px |
| ring_area_r30_px2 | normalised | 8909.82 | 8646 | 1432.67 | px^2 |
| ring_area_r60_px2 | normalised | 22488.8 | 22056 | 3166.3 | px^2 |
| in_frame_fraction_r30 | normalised | 1 | 1 | 0 | unitless |
| in_frame_fraction_r60 | normalised | 0.9972 | 1 | 0.0094 | unitless |
| ring_density_r30 | normalised | 0.0478154 | 0.0490921 | 0.00947425 | px^-1 |
| ring_density_r60 | normalised | 0.0442888 | 0.0452485 | 0.00791038 | px^-1 |
| roots_per_100px_perimeter | normalised | 4.6662 | 4.6282 | 1.0772 | per 100 px |
| ring_attached_length_r30_px | network v2 | 389.818 | 399 | 93.1051 | px |
| ring_attached_length_r60_px | network v2 | 824.182 | 828 | 220.279 | px |
| ring_length_w_r30_px | network v2 | 479.951 | 514.316 | 101.751 | px |
| ring_length_w_r60_px | network v2 | 1119.82 | 1128.31 | 211.225 | px |
| sholl_crossings_r10 | network v2 | 10.1818 | 10 | 2.6389 | count |
| sholl_crossings_r20 | network v2 | 12.6364 | 13 | 2.6181 | count |
| sholl_crossings_r30 | network v2 | 13.8182 | 13 | 3.2193 | count |
| loop_count | loops, report only | 13.0909 | 10 | 10.2123 | count |
| loop_edge_share | loops, report only | 0.3429 | 0.3495 | 0.1099 | share |
| ring_width_mean_r30_px | width | 6.8505 | 6.952 | 0.5632 | px |

Per field
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| measure | value | units |
|---|---|---|
| analysed_area_px2 | 1.02271e+06 | px^2 |
| total_skeleton_length_px | 41329 | px |
| canalicular_length_density_per_px | 0.0404112 | px^-1 |
| junction_count | 987 | count |
| junction_density_per_px2 | 0.00096508 | px^-2 |
| lacuna_count | 14 | count |

| | | |
|---|---|---|
| lacunae_per_px2 | 1.369e-05 | px^-2 |
| skeleton_component_count | 242 | count |
| mean_component_length_px | 170.781 | px |
| median_component_length_px | 16.5 | px |
| field_length_density_w_per_px | 0.0457258 | px^-1 |
| field_density_without_flagged_per_px | 0.0401152 | px^-1 |
| field_density_in_roi_per_px | | px^-1 |
| unattached_length_px | 530 | px |
| unattached_component_count | 29 | count |
| canalicular_length_density_attached_per_px | 0.0404112 | px^-1 |
| field_length_density_with_unattached_per_px | 0.0409294 | px^-1 |
| loop_count | 218 | count |
| loop_edge_share | 0.3908 | share |
| edge_count | 2142 | count |

Canalicular width, the new measure

| measure | value | units |
|---|---|---|
| width_mean_px | 7.8295 | px |
| width_median_px | 8 | px |
| width_p10_px | 4.4721 | px |
| width_p90_px | 10.198 | px |
| width_pixels_used | 38500 | count |

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel. Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every thread), and it has not been validated against any manual measurement.

Lacuna shape

| lacuna_id | area_px2 | major_axis_length_px | minor_axis_length_px | aspect_ratio | eccentricity | solidity | orientation_rad | centroid_row_px | centroid_col_px | on_border |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 510 | 36.0388 | 20.5481 | 1.7539 | 0.8215 | 0.8095 | -0.2657 | 13.3863 | 106.873 | True |
| 2 | 448 | 43.7126 | 19.8458 | 2.2026 | 0.891 | 0.6767 | 1.435 | 7.0045 | 363.067 | True |
| 3 | 1379 | 89.4664 | 21.3431 | 4.1918 | 0.9711 | 0.7971 | -1.5436 | 60.8564 | 471.593 | False |
| 4 | 1062 | 83.8665 | 19.2697 | 4.3522 | 0.9732 | 0.6923 | -1.1353 | 135.124 | 920.395 | False |
| 5 | 4132 | 136.42 | 39.4244 | 3.4603 | 0.9573 | 0.9008 | 1.5344 | 431.979 | 214.138 | False |
| 6 | 1032 | 65.9485 | 21.7201 | 3.0363 | 0.9442 | 0.8418 | -0.9507 | 441.636 | 703.582 | False |
| 7 | 2868 | 91.0398 | 40.6809 | 2.2379 | 0.8946 | 0.9166 | -1.49 | 534.719 | 477.113 | False |
| 8 | 980 | 60.8458 | 24.7024 | 2.4632 | 0.9139 | 0.7644 | -1.4596 | 557.819 | 348.526 | False |
| 9 | 2962 | 111.965 | 35.5084 | 3.1532 | 0.9484 | 0.917 | 1.4275 | 591.514 | 47.6516 | True |
| 10 | 2810 | 108.692 | 33.8251 | 3.2134 | 0.9503 | 0.9201 | 1.4475 | 714.117 | 172.09 | False |
| 11 | 1297 | 66.0375 | 26.8577 | 2.4588 | 0.9136 | 0.9134 | -1.189 | 725.743 | 851.213 | False |
| 12 | 1519 | 86.1566 | 25.1812 | 3.4215 | 0.9563 | 0.8388 | 1.3918 | 837.228 | 113.164 | False |
| 13 | 2083 | 63.4325 | 42.3049 | 1.4994 | 0.7451 | 0.9485 | -1.2565 | 905.171 | 752.206 | False |
| 14 | 2782 | 91.2948 | 39.6911 | 2.3001 | 0.9005 | 0.8762 | -1.4405 | 919.61 | 300.233 | False |

Network per lacuna

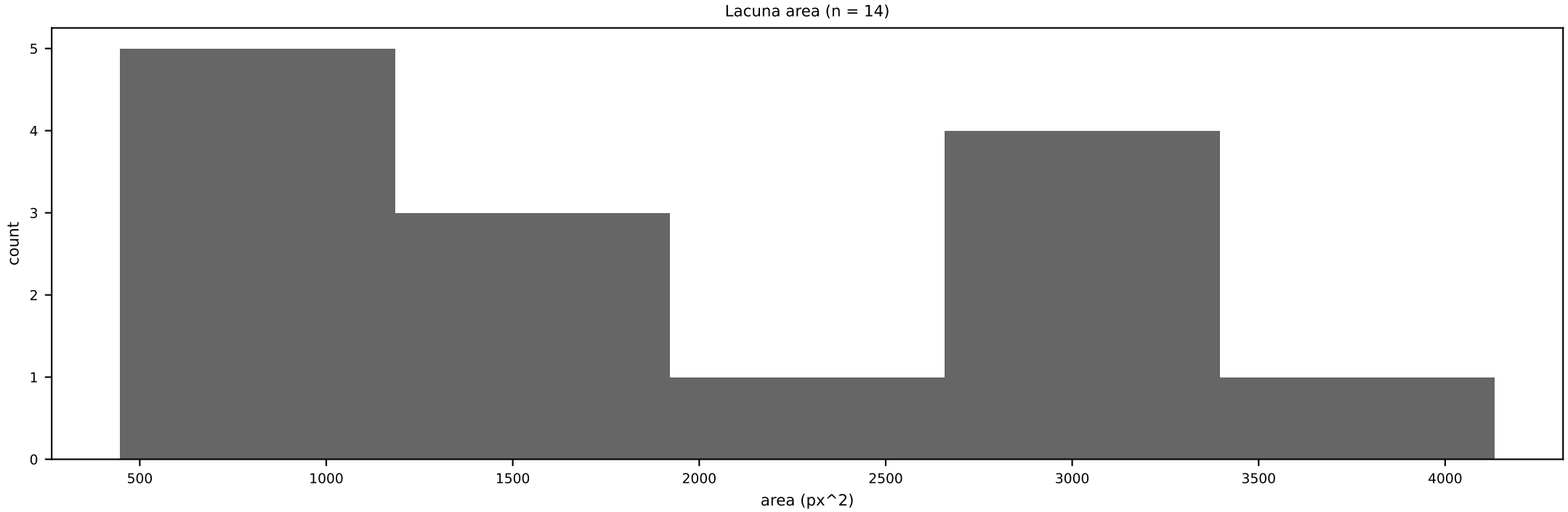
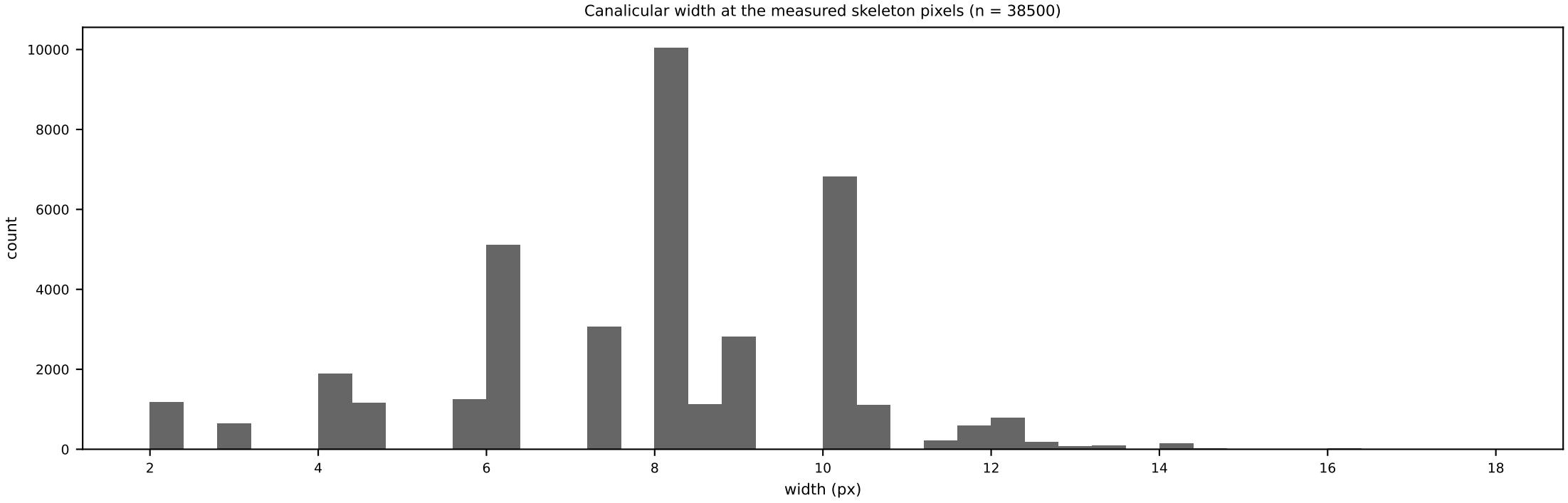
| lacuna_id | roots_count | ring_length_r30_px | ring_length_r60_px | owned_length_px | edge_count | mean_edge_length_px | ring_width_mean_r30_px |
|---|---|---|---|---|---|---|---|
| 1 | 4 | 160 | 449 | 3147.48 | 160 | 19.6717 | |
| 2 | 4 | 77 | 223 | 1381.47 | 78 | 17.7111 | |
| 3 | 8 | 400 | 825 | 1845.5 | 95 | 19.4263 | 6.0035 |
| 4 | 11 | 482 | 1109 | 5491.25 | 276 | 19.8958 | 6.952 |
| 5 | 16 | 537 | 1190 | 4775.42 | 216 | 22.1084 | 7.3889 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 6 | 5 | 421 | 1046 | 4046.57 | 198 | 20.4372 | 7.6867 |
| 7 | 13 | 465 | 998 | 4104.65 | 193 | 21.2676 | 6.8683 |
| 8 | 10 | 460 | 980 | 1999.21 | 101 | 19.7941 | 7.4533 |
| 9 | 10 | 351 | 815 | 1097.88 | 52 | 21.1131 | 7.4606 |
| 10 | 10 | 546 | 1365 | 2421.54 | 103 | 23.5101 | 7.0158 |
| 11 | 7 | 311 | 739 | 2780.9 | 107 | 25.9898 | 6.7399 |
| 12 | 14 | 442 | 1011 | 1505.18 | 68 | 22.135 | 6.0426 |
| 13 | 8 | 267 | 790 | 1391.31 | 65 | 21.4048 | 6.9734 |
| 14 | 12 | 323 | 827 | 2142.01 | 99 | 21.6365 | 6.2309 |

Normalised per lacuna

| lacuna_id | perimeter_px | ring_area_r30_px2 | ring_area_r60_px2 | in_frame_fraction_r30 | in_frame_fraction_r60 | ring_density_r30 | ring_density_r60 | roots_per_100px_perimeter |
|---|---|---|---|---|---|---|---|---|
| 1 | 105.154 | 3701 | 10161 | 0.6475 | 0.597 | 0.0432316 | 0.0441886 | 3.8039 |
| 2 | 132.847 | 3699 | 7686 | 0.6031 | 0.5656 | 0.0208164 | 0.0290138 | 3.011 |
| 3 | 219.681 | 8976 | 20715 | 1 | 0.9689 | 0.0445633 | 0.0398262 | 3.6416 |
| 4 | 210.244 | 8511 | 22591 | 1 | 1 | 0.0566326 | 0.0490903 | 5.232 |
| 5 | 335.421 | 12178 | 29932 | 1 | 1 | 0.0440959 | 0.0397568 | 4.7701 |
| 6 | 167.238 | 7485 | 20582 | 1 | 1 | 0.0562458 | 0.0508211 | 2.9898 |
| 7 | 243.522 | 9472 | 22056 | 1 | 1 | 0.0490921 | 0.0452485 | 5.3383 |
| 8 | 167.924 | 7349 | 17555 | 1 | 1 | 0.0625935 | 0.0558246 | 5.9551 |
| 9 | 282.894 | 8678 | 19332 | 0.813 | 0.7462 | 0.0404471 | 0.0421581 | 3.5349 |
| 10 | 263.48 | 10172 | 24552 | 1 | 1 | 0.0536768 | 0.0555963 | 3.7954 |
| 11 | 174.51 | 7570 | 20770 | 1 | 1 | 0.0410832 | 0.0355802 | 4.0112 |
| 12 | 210.167 | 8646 | 22189 | 1 | 1 | 0.0511219 | 0.0455631 | 6.6614 |
| 13 | 185.823 | 7980 | 21610 | 1 | 1 | 0.0334586 | 0.0365571 | 4.3052 |
| 14 | 259.279 | 9669 | 24825 | 1 | 1 | 0.0334057 | 0.0333132 | 4.6282 |

Network v2 per lacuna

| lacuna_id | ring_attached_length_r30_px | ring_attached_length_r60_px | ring_length_w_r30_px | ring_length_w_r60_px | sholl_crossings_r10 | sholl_crossings_r20 | sholl_crossings_r30 |
|---|---|---|---|---|---|---|---|
| 1 | 145 | 372 | 175.468 | 498.019 | 5 | 5 | 6 |
| 2 | 70 | 158 | 87.0122 | 245.024 | 2 | 1 | 3 |
| 3 | 314 | 530 | 444.291 | 911.34 | 10 | 12 | 15 |
| 4 | 477 | 1084 | 554.588 | 1272.64 | 13 | 13 | 12 |
| 5 | 508 | 942 | 597.203 | 1326.49 | 10 | 17 | 15 |
| 6 | 399 | 841 | 493.446 | 1207.89 | 11 | 13 | 13 |
| 7 | 439 | 828 | 524.475 | 1128.31 | 12 | 14 | 12 |
| 8 | 448 | 951 | 520.162 | 1099.67 | 11 | 13 | 15 |
| 9 | 259 | 547 | 400.191 | 919.21 | 9 | 12 | 11 |
| 10 | 499 | 1228 | 611.546 | 1536.79 | 15 | 16 | 21 |
| 11 | 267 | 627 | 351.664 | 853.394 | 6 | 12 | 10 |
| 12 | 384 | 825 | 514.316 | 1148.35 | 9 | 12 | 17 |
| 13 | 257 | 643 | 306.179 | 908.365 | 8 | 9 | 11 |
| 14 | 296 | 567 | 361.593 | 924.826 | 7 | 8 | 11 |

682_z23: distributions, pre-validation, px